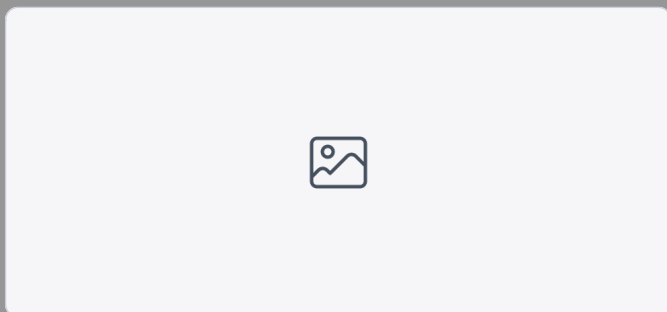


Robot Kinematics.

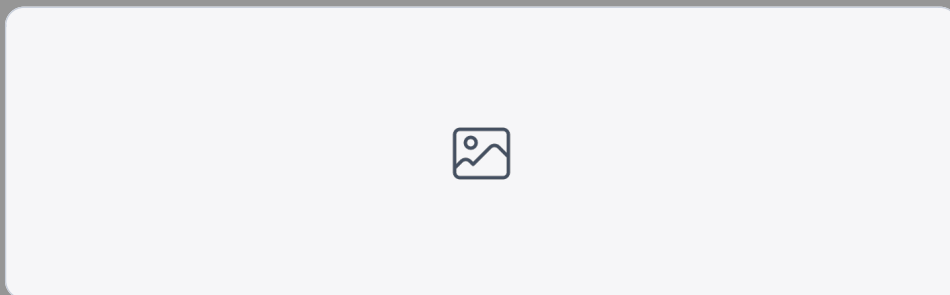
> planar 2-link arm:

Base — $\angle 1$ — Joint — $\angle 2$ — End Effector
 $\theta 1$ $\theta 2$



the second link's
world angle:
 $\theta 1 + \theta 2$

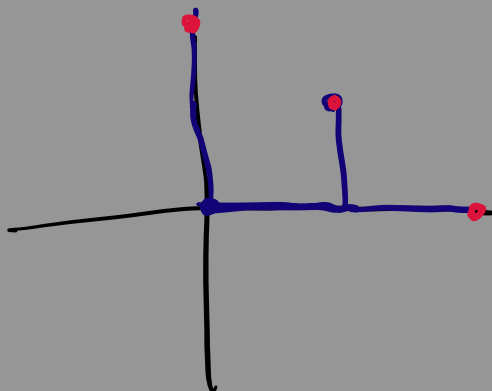
two 100mm links



1. (200, 0)

2. (0, 200)

3. (100, 100)



> end effector is the sum of two link vectors

first joint pos: $x_1 = L_1 \cdot \cos(\theta_1)$
 $y_1 = L_1 \sin(\theta_1)$

second link's abs/world angle: $\theta_1 + \theta_2$

↳ offset from the first joint:

$$\Delta x_2 = L_2 \cdot \cos(\theta_1 + \theta_2)$$

$$\Delta y_2 = L_2 \cdot \sin(\theta_1 + \theta_2)$$

+

$$x = L_1 \cdot \cos(\theta_1) + L_2 \cdot \cos(\theta_1 + \theta_2)$$

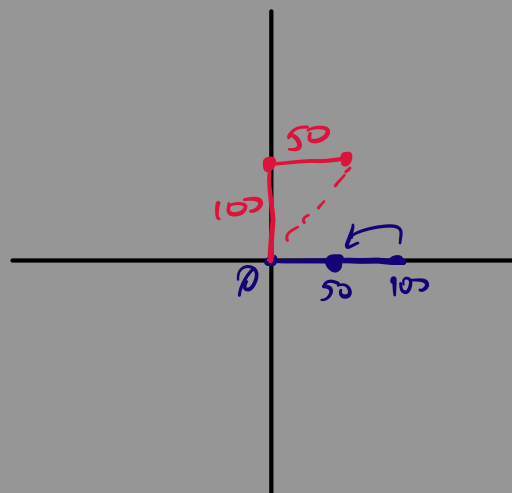
$$y = L_1 \cdot \sin(\theta_1) + L_2 \cdot \sin(\theta_1 + \theta_2)$$

$$L_1 = 100\text{mm} \quad ? \quad L_2 = 50\text{mm}$$



$$1. (50, 0)$$

$$2. (50, 100)$$



> Joint space

describes the robot using its joint angles

$$(\theta_1, \theta_2)$$

> Task/Cartesian Space

describes the end effector using position

$$(x, y)$$

$\Rightarrow$ forward kinematics maps

Joint $\rightarrow$ Task/Cartesian

$$(\theta_1, \theta_2) \rightarrow (x, y)$$

$\rightarrow$ inverse kinematics
can have
multiple
solutions

! different

Joint configs
can reach the
same cartesian
point

(but elbows pos.
may differ)



1. input: θ_1, θ_2 + fixed robot dim L_1, L_2 ; output: end-effector (x, y)
2. Joint space
3. no, end may be same but their elbows can differ

this is the same chaining from prev.

$$T_{base \leftarrow end} = T_{base \leftarrow link1} \cdot T_{link1 \leftarrow link2}$$

$$T = \begin{bmatrix} R_{2 \times 2} & t_{2 \times 1} \\ 0 & 1 \end{bmatrix}$$



1. 3×3
2. becu2 the rotations compose
3. $T_{link1 \leftarrow link2}$

→ forward kinematics doesnot have ambiguity
inverse kinematics does

Inverse Kinematics

> reject impossible targets, before calculating opks

$$\begin{aligned} \angle 1: 100 \text{ mm} \\ \angle 2: 50 \text{ mm} \end{aligned} \Rightarrow \begin{aligned} &\text{reachable distance:} \\ &50 \leq r \leq 150 \\ &r = \sqrt{x^2 + y^2} \end{aligned}$$

once it's reachable:

$$\cos(\theta_2) = \frac{(x^2 + y^2 - L_1^2 - L_2^2)}{2 L_1 L_2}$$

$$= \theta_2 = \pm \arccos(\cos \theta_2)$$

↙ elbow up
↘ elbow down

EXAMPLE

$$L_1 = 100$$

$$\text{Target} = (100, 50)$$

$$L_2 = 50$$

$$x^2 + y^2 = 12500$$

$$\text{numerator} = 12500 - 100^2 - 50^2$$

$$\cos \theta_2 = 0$$

$$\theta_2 = \pi/2, -\pi/2$$

$$\theta_1 = \underbrace{\text{atan2}(y, x)}_{\substack{\text{dir. from} \\ \text{base to} \\ \text{target}}} - \underbrace{\text{atan2}(L_2 \cdot \sin(\theta_2), L_1 + L_2 \cos(\theta_2))}_{\substack{\text{angular offset} \\ \text{between link 1} \\ \text{and target dir.}}}$$

$$\text{atan2}(50, 100) = 26.565^\circ$$

for $\pi/2$

$$L_2 \sin \theta_2 = 50$$

$$L_1 + L_2 \cos \theta_2 = 100$$

for $-\pi/2$

$$L_2 \sin \theta_2 = -50$$

$$L_1 + L_2 \cos \theta_2 = 100$$

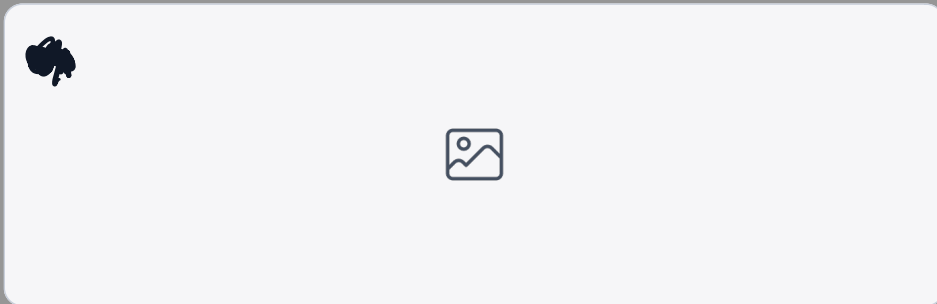
$$\theta_1 = 56.5^\circ$$

$$\approx 26,565^\circ$$

$$\approx -26,565^\circ$$

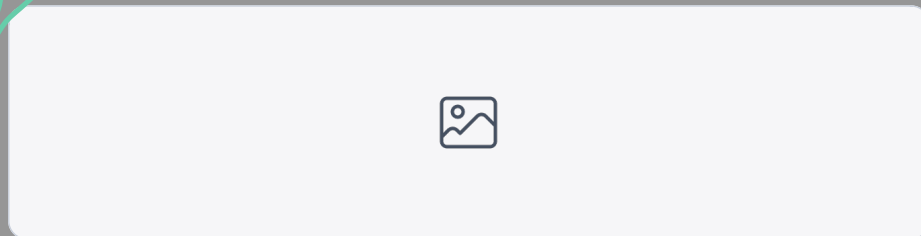
$\Rightarrow 0$

$$\Rightarrow 53,13$$



target position $\xrightarrow{IK}$ Joint angles $\xrightarrow{FK}$ calculated positions

A/



✓ verifies

$$B/ \quad 100 \times \cos 53,13 + 50 \times \cos(-36,87) = 100$$

$$100 \times \sin 53,13 + 50 \times \sin(-36,87) = 50$$

✓ verifies